AI infrastructure and distributed systems

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AI infrastructure and platform engineer with 12 years building and operating production distributed systems. At Apple, I co-owned an internal enterprise e2e private AI serving platform and owned model-weight tooling and Solr infrastructure. My work connects architecture, hands-on automation, performance engineering and incident response.

01 Enterprise GenAI platform

Apple · Contract via Mphasis · Apr 2024–Aug 2026

02 Cyber telemetry & distributed ingestion

Cloudwick · Cyber Data Lake / Amorphic · Nov 2016–Dec 2018

03 Mission-critical cloud migration

Cloudwick · Client: Freddie Mac · Jan 2019–Aug 2023

04 Cloud automation, identity & recovery

Cloudwick · Client: Freddie Mac · Jan 2019–Aug 2023

Sanitized reconstructions and career-reported outcomes. Full case studies and supporting diagrams are available on the website.

CASE 01 / AI INFRASTRUCTURE

Enterprise GenAI platform

Apple | Contract via Mphasis | Apr 2024–Aug 2026

I co-owned platform engineering, GPU/Kubernetes infrastructure, ingestion and inference GitOps onboarding, and observability for an internal enterprise e2e private AI serving platform. I owned model-weight serving and onboarding tooling, Solr infrastructure and performance engineering, and Mac document-conversion infrastructure. Elasticsearch and Milvus were shared ownership.

15,000+ collections

Project embeddings across five Solr clusters in one region; at least 90% active

7.0s → 1.9s

Approximate vector-query latency

$500K+

Combined Solr + Mac infrastructure savings

What I owned

• MODEL-WEIGHT TOOLING: Owned serving, onboarding, rapid prototyping and decommissioning tooling.

• SEARCH & VECTOR INFRASTRUCTURE: Owned Solr; co-owned Elasticsearch and Milvus: provisioning, tuning, tooling and IaC.

• KUBERNETES PLATFORM OPERATIONS: Co-owned ingestion/inference and GPU Kubernetes lifecycle, GitOps and operations.

• PRODUCTION RELIABILITY: Co-owned observability and incident response across network, application, queues and storage.

Engineering decisions

Separate application and model infrastructure

The architecture separates the ingestion/inference services from GPU model hosting. This lets application and model deployments have distinct capacity and runtime configurations.

Make model artifacts part of deployment

Weight staging, integrity checks, shared volumes, health probes and runtime values are first-class deployment steps. Configuration covers precision, context limits, memory budgets, tensor parallelism and LoRA serving.

Treat retrieval as a systems problem

I worked across JVM/GC, OS settings, cache behavior, query patterns and indexes. The reported latency improvement is a vector-query result, not an LLM generation benchmark.

Scope: Sanitized logical architecture, with personal and team ownership distinguished. Configured worker and replica counts describe desired configuration rather than a live fleet inventory. Model names describe platform serving scope, not training ownership. Latency figures refer to vector queries; no LLM throughput, TTFT or GPU-utilization improvement is claimed.

CASE 01 / TOOLING & PRODUCTION

Hands-on AI platform engineering

Go & platform automation

Personally wrote or modified the model-weight checksum CLI and Kubernetes Kafka topic/user resource generator. Built lifecycle automation in Python, Terraform, CloudFormation and Ansible.

GPU & model serving

Co-owned GPU Kubernetes setup, upgrades, patching and operations on NVIDIA V100, A10G, L40S and A100 GPUs. Owned model-weight onboarding and serving tooling with Helm, Argo CD and vLLM.

Model platform

Apple foundation models; Qwen3.5-397B-A17B-FP8; Qwen3-32B; Mixtral-8x22B-Instruct-v0.1; gpt-oss-120b; Pixtral-12B-2409.

Production investigation

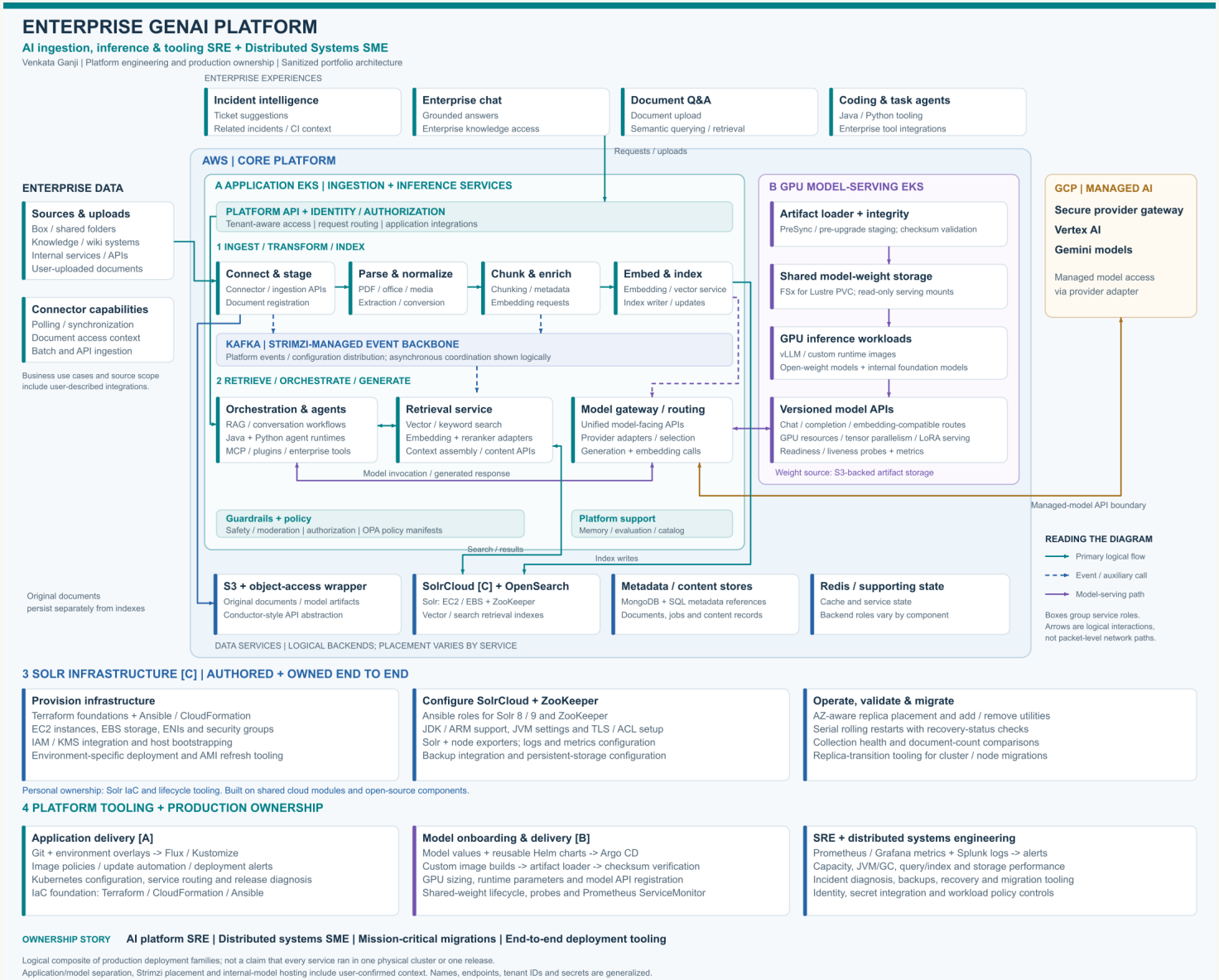
Co-owned observability and incident response. Investigated network PCAP, p95/p99 latency, request/ingestion queues and Kubernetes PVC failures.

Scale & configuration

Supported 15,000+ embedding collections across five Solr clusters in one region, at least 90% active. Production configurations specified 226 application workers across two AWS regions and 105 document-processing replicas; these are configured counts, not live fleet inventory.

Model identifiers reflect serving-platform scope. No Apple foundation-model version or model-training ownership is asserted.

Enterprise GenAI platform



Explore the full-size diagram, system flow, results and supporting views online →

CASE 02 / DISTRIBUTED INGESTION

Cyber telemetry & distributed ingestion

Cloudwick | Cyber Data Lake / Amorphic | Nov 2016–Dec 2018

I owned the three-rack cyber data lake infrastructure end to end: hardware and network integration, operational tooling and performance tuning. I connected PCAP, NetFlow, application logs and firewall events to a correlated distributed-data platform.

3 racks

Capture and integration test bed

40 Gb/s

Packet-capture technology tuned

4 source types

PCAP, flow, app and firewall telemetry

What I owned

• Implemented collection of PCAP, NetFlow, application logs and firewall events to support a correlated cyber data lake.

• Owned Supermicro servers, high-throughput NICs, IXIA and ToR / supervisor-switch integration in the three-rack capture/replay environment; tuned 40 Gb/s capture technology.

• Owned operational and performance tooling across capture, hardware and distributed data; engineered Corelight/IXIA integrations and operated Hadoop, Kafka, Spark, Solr, HBase and Elasticsearch.

• Built Prometheus monitoring for collector agents and automated cloud/on-premises deployment using CloudFormation, Ansible, Jenkins, GitLab, Kickstart, Puppet and Python APIs.

Engineering decisions

Separate functional tests from performance tests

A functionally correct collector can still be limited by disk, broker capacity or the capture path. The test environment made those dimensions visible instead of treating every failure as an application problem.

Keep capture and management paths distinct

The multi-rack design separates the data-transfer network from server-management access. Capture/replay and security integrations can be reasoned about independently from routine host operations.

Observe the collector as well as the cluster

Prometheus monitoring at the collector layer complements distributed-platform operations. This gives the team visibility before data reaches the search or analytics tier.

Scope: The walkthrough diagrams are simplified redraws; low-resolution previews of the supplied originals are available separately. Gateway capacities and test configurations are not promoted to production benchmarks. The separate Amorphic cloud reference is dated June 2019 and credits a team author; it is included only as product context, not as evidence that I personally designed that entire later architecture. Telemetry here does not mean OpenTelemetry.

CASE 02 / ARCHITECTURE

Cyber telemetry & distributed ingestion

Cyber telemetry and distributed ingestion

Logical platform roles across capture, collection, storage and investigation

PACKET PATH

Mirrored network traffic
Out-of-band source traffic

→

Packet broker
Capture distribution

→

Capture systems
40 Gb/s technology test bed

→

Packet storage
Capture storage / HDFS

EVENT PATH

Logs + flow telemetry
App / firewall / NetFlow

→

Collectors + aggregator
Telemetry collection

→

Kafka
Brokered ingestion

→

HDFS + analytics
Distributed persistence

CROSS-CUTTING CAPABILITIES

Distributed processing
Spark / Hadoop

Search + investigation
Solr / HBase / Elasticsearch

Collector visibility
Prometheus monitoring

Repeatable deployment
Ansible / cloud + host tools

Rows group platform capabilities; they do not assert one pipeline for every data source.

Explore the full-size diagram, system flow, results and supporting views online →

CASE 03 / DATA MIGRATION

Mission-critical cloud migration

Cloudwick | Client: Freddie Mac | Jan 2019–Aug 2023

I delivered solutions architecture, software engineering and production operations across a nearly four-year Freddie Mac engagement. I led three mission-critical migrations, designed the Hadoop/Solr-to-S3/search transition, and led Spark migration development through production cutover.

3 migrations
Mission-critical programs led in the engagement

3–8 engineers
Approximate teams technically led

HDFS → S3
Document-storage modernization

What I owned

• Led three mission-critical migrations across distributed-data and live application systems during the engagement, coordinating architecture, automation, data movement, performance testing and production transition.

• Designed and productionized a document-platform migration from Cloudera Hadoop/Solr to AWS Elasticsearch/OpenSearch, ELK and S3.

• Led development of Spark migration jobs moving Solr data into Elasticsearch/ELK and HBase data into S3.

• Built reusable IaC and operations automation; designed Databricks, AWS Glue, Talend and S3 ingestion frameworks and Python/Airflow workflows; supported performance testing and production transitions.

Engineering decisions

Separate transport from transformation

DistCp can address bulk transfer, while archive extraction and target naming/mapping require additional work. The design makes those responsibilities explicit.

Keep document and search migrations coordinated

S3 objects and search records serve different roles. The migration program covered both, so the target architecture could support application behavior as well as storage.

Include operations in the migration scope

Provisioning, performance testing, snapshots and recovery were part of the engineering work. A cloud cutover needed an operating model as well as a target diagram.

Scope: The HAR diagram presents alternatives. It does not by itself establish the selected route for every workload, total bytes migrated, zero downtime or a benchmark. The leadership and delivery claims come from my career record; the drawing explains the technical approach.

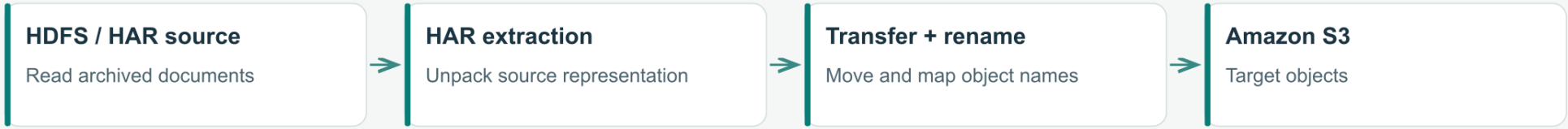
CASE 03 / ARCHITECTURE

Mission-critical cloud migration

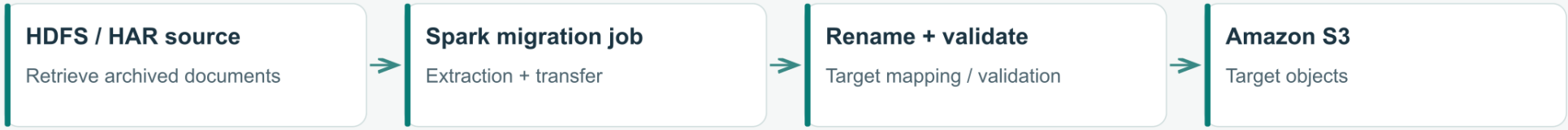
Hadoop archives to cloud object storage

Migration approaches shown in the design; separate from the search-data migration

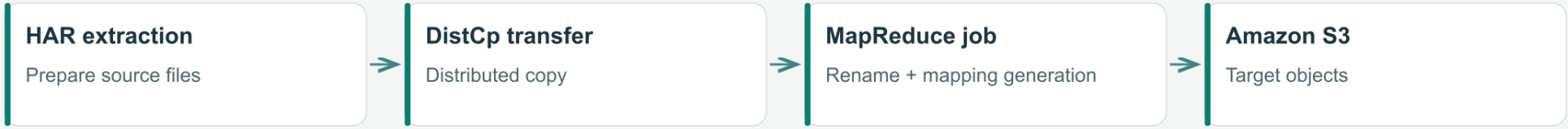
STAGED APPROACH



CUSTOM SPARK APPROACH



DISTCP + MAPREDUCE APPROACH



The source drawing compares approaches; it does not establish one chosen route for every workload.

Explore the full-size diagram, system flow, results and supporting views online →

CASE 04 / PLATFORM ENGINEERING

Cloud automation, identity & recovery

Cloudwick | Client: Freddie Mac | Jan 2019–Aug 2023

I designed and productionized cloud infrastructure for a document and search platform, including proxy access, federated authentication, reusable IaC and hybrid-cloud disaster recovery. The work joined application requirements with security and operational constraints.

~15 min

RPO for the delivered DR environment

~2 hours

RTO for the delivered DR environment

4 environments

Federated authentication scope

What I owned

• Designed proxy, load-balancing and autoscaling infrastructure supporting the document platform's move to AWS.

• Developed reusable Terraform, CloudFormation and Ansible provisioning/landing-zone workflows covering compute, search, networking, IAM, secrets, encryption and cross-region storage.

• Designed federated authentication using PingFederate, Vault, AWS IAM/STS, ADFS and SAML across four environments; partnered with InfoSec on on-premises-to-S3 connectivity.

• Built and productionized a hybrid-cloud DR environment using cross-region replication, S3, Airflow, Python and Elasticsearch/ELK, delivered within six months.

Engineering decisions

Bridge the legacy application boundary

The proxy tier provides an integration point between the existing application and cloud services. The design also spreads proxies across availability zones and uses autoscaling.

Treat identity as architecture

Directory integration, federation and temporary AWS credentials influence how applications reach the data plane. They were designed alongside network paths, not added as an afterthought.

Make recovery and IaC reinforce each other

Reusable provisioning defines the target resources; backup/replication provides recoverable state. The architecture distinguishes those responsibilities, including an on-demand recovery footprint.

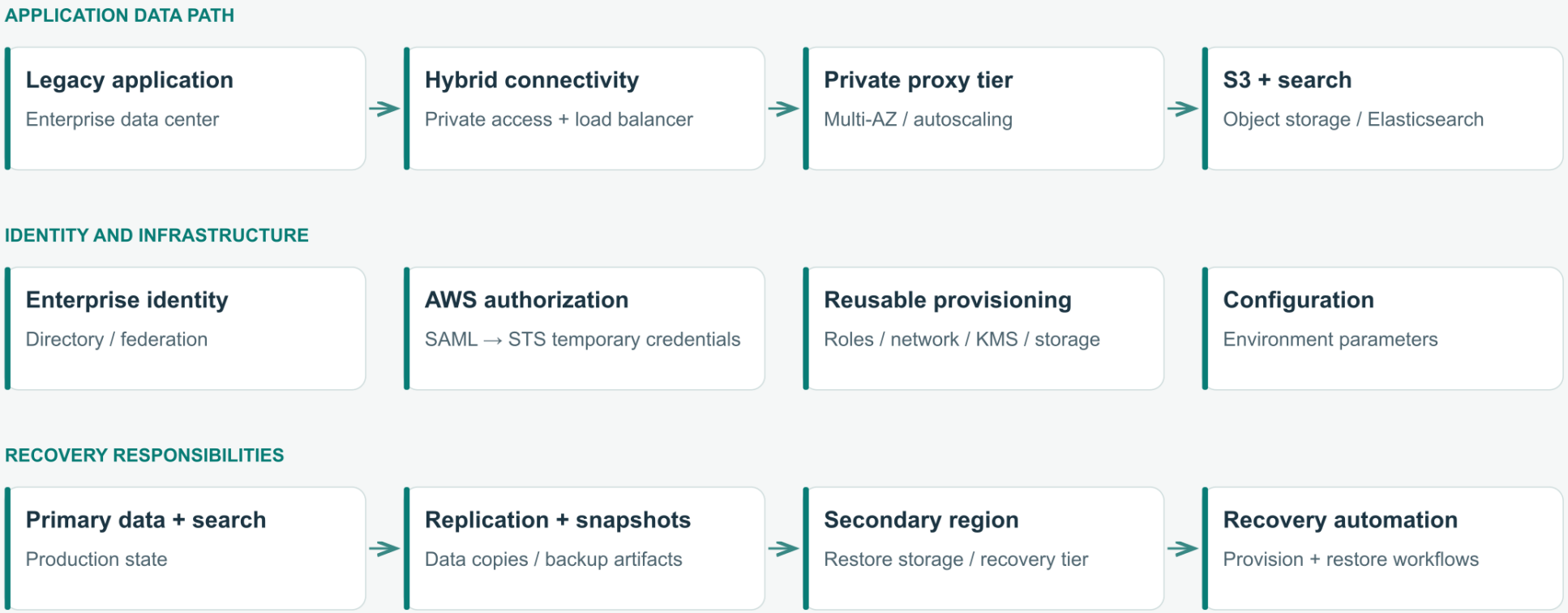
Scope: Drawings represent different releases and environments; the portfolio shows their relationship without implying that every identity mechanism runs in the same request path. Recovery metrics are career-reported results for the described DR environment, not a universal service guarantee.

CASE 04 / ARCHITECTURE

Cloud automation, identity & recovery

Hybrid cloud document and search platform

Application access, cloud foundations and recovery shown as distinct responsibilities



Logical composite of releases. DR results: approximately 15-minute RPO and 2-hour RTO for the delivered environment.

Explore the full-size diagram, system flow, results and supporting views online →